

LLM-based pathway interpretation of SynergyLM: a pre-registered test of biological correctness

SynergyLM extension, technical report

1 October 2026

Summary

- **Question.** Do SynergyLM’s per-gene attributions make an LLM agent’s mechanistic explanations of a drug combination more biologically correct than the same agent reasoning over the same knowledge graph (KG) without attributions? “Correct” means the named Reactome pathway is a cell-line-specific CRISPR dependency (DepMap) in the cell line being explained.
- **Setup.** 250 synergistic and 250 antagonistic drug-pair–cell-line items; two SynergyLM variants (multi-omics, L1000); an attribution-guided agent and a KG-only agent, both run locally on Qwen/Qwen3-32B-AWQ with every claim checked by a mechanical verifier; deterministic no-LLM selectors as exploratory controls. 1,500 LLM runs.
- **Result: null.** The primary hypothesis (H1) fails for both variants. Multi-omics: mean difference +0.03 SD (95% CI -0.17 to $+0.23$, $n = 93$). L1000: +0.22 SD (CI -0.09 to $+0.57$, $n = 20$, underpowered). The secondary hypotheses H2 (synergy vs antagonism) and H3 (L1000 vs multi-omics) are also null.
- **The LLM is not the bottleneck.** The guardrails worked (92–99% of claims valid on the first try) and attributions were stable across seeds. But choosing the pathway from the attributions *without* an LLM is no better than choosing a random convergent pathway, and the LLM agent does not improve on that selector. The attributions carry no detectable pathway-level signal about dependency. A larger or paid LLM would not change this.
- **Likely cause.** SynergyLM’s omics inputs behave as a cell-line identifier (59 cell profiles, 27,606 features), so gene-level attributions describe which cell line it is rather than mechanism. Later experiments confirmed this directly (Section 5).

1 Question and hypotheses

Table 1: Pre-registered hypotheses (plan of 2026-09-29, fixed before any agent run).

ID	Hypothesis	Role
H1	For synergistic items, the pathway named by the attribution-guided agent has higher cell-line-specific dependency than the pathway named by the KG-only agent. Tested separately for each model variant.	Primary
H2	The attribution agent’s advantage is larger for synergistic than for antagonistic items.	Secondary
H3	The advantage is larger with the L1000 variant (drug-side gene signal) than with the multi-omics variant.	Secondary

Falsification rule. If H1 fails for both variants, the attributions add no biological correctness beyond the KG and the LLM’s prior. This was declared a reportable negative result before the study ran.

2 Methods

Models. SynergyLM (ACM BCB 2026) predicts the ZIP synergy score of a drug pair in a cell line from frozen MolFormer drug embeddings and cell-line omics. Two variants were explained: *multi-omics* (mRNA 23,808, protein 3,171 and miRNA 627 features fused by attention) and *L1000* (mRNA plus each drug’s L1000 landmark signature, 978 genes). Each variant was trained with three seeds on TDC’s random split (ZIP target, published hyperparameters).

Attributions. Integrated Gradients (64 steps) with respect to the raw inputs, against the mean training profile as baseline, so an attribution means “what is distinctive about this cell line or drug”. For each channel, $|IG|$ was summed over each Reactome pathway with at least 5 genes in the channel and compared with 10,000 size-matched random gene sets drawn from the channel’s Reactome-annotated genes. Benjamini–Hochberg FDR was computed within each channel. A pathway is *attribution-supported* if its FDR is below 0.1 in at least 2 of 3 seeds. miRNA cannot enter a pathway explanation.

Knowledge graph. PrimeKG drug–protein target edges, Reactome (all levels, human, 10–300 genes: 1,733 pathways), and GO biological process as non-claimable context. KEGG was omitted pending licence terms. 100 of 129 drugs have a protein target; drugs without one (platinum compounds, most alkylators) make an item ineligible.

Agent. Qwen/Qwen3-32B-AWQ (Apache-2.0), served locally with vLLM on one 40 GB A100 (32,768-token context). Thinking mode was on, with the model card’s sampling settings (temperature 0.6, top- p 0.95, top- k 20). A fixed per-request seed made runs reproducible, and each run had at most 25 turns. The agent receives the drug pair, the cell line and KG tools, and must return strict JSON: a primary Reactome pathway, up to three ranked pathways, and for each a KG path target(A) $\rightarrow$ pathway $\leftarrow$ target(B). A mechanical verifier outside the LLM checks that every edge exists in the KG and, in the attribution condition, that the pathway is attribution-supported. Rejected outputs get one retry, and the agent may abstain.

Conditions.

- *Attribution agent* (one per variant): the agent plus a tool listing the item’s admissible attribution-supported pathways.
- *KG-only agent*: the same agent without attributions. It runs once per item because it does not depend on the variant.
- Exploratory, not pre-registered:
 - *attribution selector*: the highest-scoring admissible pathway, chosen with no LLM;
 - *random*: the exact expected score of a uniform choice among the pair’s convergent pathways.

Outcome. For pathway P and cell line C , with Chronos gene effects from DepMap 24Q4 and at least 10 screened genes in P :

$$\text{dep}(P, c) = \frac{1}{|P|} \sum_{g \in P} \text{Chronos}(g, c), \quad \text{spec}(P, C) = - \frac{\text{dep}(P, C) - \text{mean}_c \text{dep}(P, c)}{\text{sd}_c \text{dep}(P, c)}.$$

Higher means P is more selectively essential in C than in other lines, which removes the advantage of pathways that are essential everywhere. Differences in spec are in standard-deviation (SD) units.

Sample. From the random-split test set:

- *Synergy stratum*: observed ZIP in the top decile and predicted in the top quintile by both variants.
- *Antagonism stratum*: the mirror image (bottom decile, bottom quintile).

Filters:

- both drugs have a KG target and at least one convergent pathway;
- neither drug uses a proxy L1000 profile;
- the cell line has DepMap CRISPR data.

Caps: at most 8 items per cell line and 3 per drug pair. 250 items were sampled per stratum.

Analysis.

- **H1**: per-item difference $\Delta = \text{spec}_{\text{attr}} - \text{spec}_{\text{KG}}$ on the synergy stratum, over items where both agents made a verified claim. Tests:
 - a linear mixed model $\Delta \sim 1 + (1|\text{cell line}) + (1|\text{drug pair})$, Holm-corrected across the two variants;
 - a 95% cluster-bootstrap CI that resamples cell lines (2,000 resamples);
 - the paired effect size d_z ;
 - a Wilcoxon signed-rank test.
Reported p -values are two-sided. The pre-registered one-sided test gives the same verdict (L1000: one-sided $p = 0.067$, Holm 0.13).
- **H2**: the condition $\times$ stratum interaction in a mixed model with crossed random intercepts for cell line, drug pair and item.
- **H3**: the paired difference $\Delta_{\text{L1000}} - \Delta_{\text{multi-omics}}$ on shared items.

3 Pre-registered gates

Table 2: Gates checked before (G0, G1) and during (G2) the agent runs.

Gate	Criterion	Observed	Outcome
G0 data	≥ 32 of 59 cell lines in DepMap CRISPR and ≥ 250 eligible synergy items	41 lines; 1,207 synergy and 602 antagonism items eligible	Pass
G1 stability	Median seed-to-seed Spearman of pathway scores ≥ 0.5	0.81 (multi-omics), 0.66 (L1000)	Pass
G2 verifier, pilot 1	Attribution-agent claims valid on first try ≥ 0.80 (20 items, 40 runs)	0.60; most errors claimed non-admissible pathways; one context overflow	Fail
G2 verifier, pilot 2	Same, after the tool was restricted to admissible pathways	0.95	Pass

A power shortfall was known before the primary analysis. Only 98 of 250 synergy items (multi-omics) and 21 of 250 (L1000) had at least one admissible attribution-supported pathway, against the 240 items the power analysis assumed. The imputed sensitivity analysis keeps all 250 items. The median item had 184 supported pathways in the multi-omics cell channel and 7 in the L1000 channels (144 of 500 L1000 items had none). miRNA carried a median 34% of the multi-omics attribution mass, which no pathway explanation can use.

4 Results

4.1 Agent behaviour

Table 3: Run outcomes per condition (250 runs each). Accepted: a verified claim; abstained: no admissible pathway claimed; rejected: failed verification after the retry. First-try valid: the fraction of first submissions passing the verifier.

Condition	Stratum	Accepted	Abstained	Rejected	First-try valid	Turns	Output tokens
Attribution agent (multi-omics)	antagonism	0.364	0.624	0.008	0.928	4.4	2,733
Attribution agent (multi-omics)	synergy	0.388	0.604	0.004	0.936	4.4	2,552
Attribution agent (L1000)	antagonism	0.036	0.952	0.000	0.988	4.4	2,534
Attribution agent (L1000)	synergy	0.084	0.916	0.000	0.992	4.3	2,344
KG-only agent	antagonism	0.992	0.004	0.004	0.948	3.2	2,548
KG-only agent	synergy	0.968	0.008	0.012	0.916	3.2	2,337

The KG-only agent made a verified claim for 97–99% of items. The attribution agents abstained for about 60% (multi-omics) and 92–95% (L1000) of items, which matches the share of items with no admissible pathway. Verifier rejections were rare (at most 1.2%). Neither refusals nor runs hitting the turn limit occurred.

4.2 Primary hypothesis H1

Table 4: H1: attribution-guided agent minus KG-only agent, synergy stratum. Δ in SD units of dependency specificity; CI by cluster bootstrap over cell lines; p two-sided. Holm-adjusted p_{mixed} for the primary rows: 0.96 (multi-omics), 0.27 (L1000).

Comparison	n items	Cell lines	Mean Δ	95% CI	d_z	p_{mixed}	p_{Wilcoxon}
<i>Primary: top-1 pathway, items where both agents made a verified claim</i>							
Multi-omics	93	32	+0.032	[−0.17, +0.23]	+0.04	0.96	0.57
L1000	20	17	+0.218	[−0.09, +0.57]	+0.30	0.13	0.17
<i>Sensitivity: abstentions scored as 0 (all 250 items)</i>							
Multi-omics	250	41	+0.008	[−0.18, +0.19]	+0.01	0.72	0.40
L1000	250	41	+0.022	[−0.20, +0.21]	+0.02	0.75	0.18
<i>Secondary: mean over the top-3 pathways</i>							
Multi-omics	93	32	+0.124	[−0.03, +0.29]	+0.19	0.26	0.17
L1000	20	17	+0.144	[−0.21, +0.56]	+0.16	0.41	0.62

H1 is not supported for either variant (Table 4, Figure 1). For multi-omics the effect is essentially zero ($d_z = 0.04$), and its CI excludes effects larger than about +0.23 SD. For L1000 the point estimate is larger (+0.22 SD), but it rests on 20 items from 17 cell lines and its CI includes zero. Scoring abstentions as zero (all 250 items) or using the top-3 pathways does not change the verdict.

4.3 Secondary hypotheses H2 and H3

Neither is supported (Table 5). For H3 the point estimate even runs the other way (L1000 advantage smaller than multi-omics by 0.07 SD), but it rests on only 12 shared items.

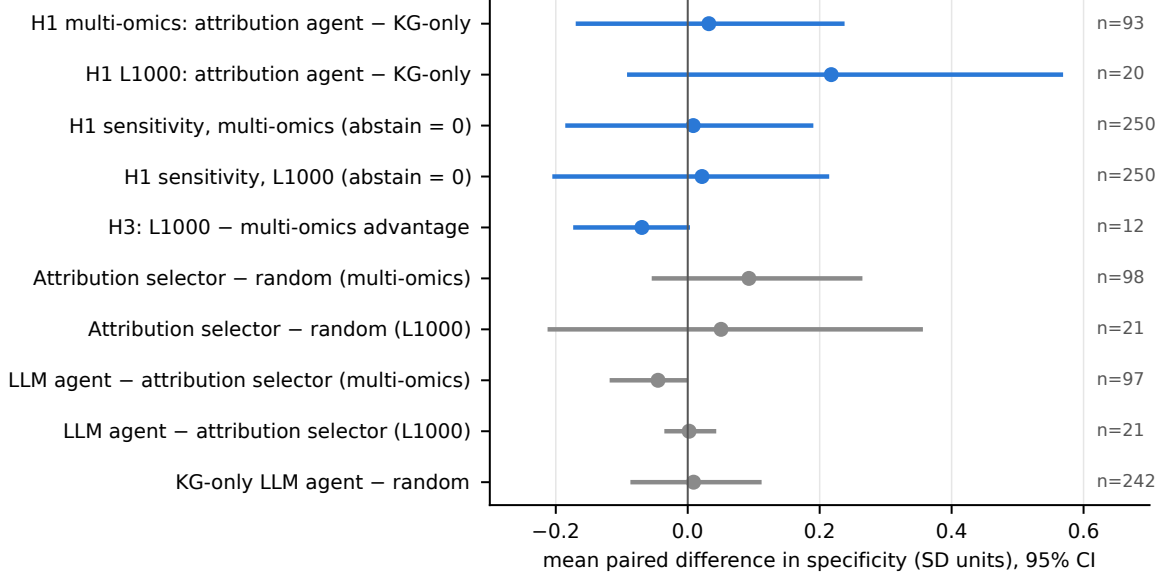


Figure 1: All paired comparisons on the synergy stratum. Blue: pre-registered (H1, H3); grey: exploratory. Points are mean differences in dependency specificity (SD units) with 95% cluster-bootstrap CIs. Every interval contains zero except the LLM-agent-minus-selector comparison (multi-omics), which favours the selector.

Table 5: H2 (does the attribution advantage differ between synergy and antagonism?) and H3 (is it larger for L1000?).

Hypothesis	Variant	<i>n</i>	Estimate	SE / 95% CI	<i>p</i>
H2: attribution $\times$ synergy interaction	Multi-omics	678	+0.058	0.101	0.57
H2: attribution $\times$ synergy interaction	L1000	520	+0.339	0.270	0.21
H3: $\Delta_{L1000} - \Delta_{\text{multi-omics}}$	both	12	-0.070	[-0.17, +0.00]	0.12

4.4 Where the signal is lost: attributions vs the LLM (exploratory)

Table 6 separates the two places a signal could enter.

1. **The attributions.** Selecting the pathway by attribution score is not reliably better than a random convergent pathway (+0.09 SD, $p = 0.29$, multi-omics; +0.05 SD, $p = 0.53$, L1000).
2. **The LLM’s choice.**
 - With attributions, the agent does slightly *worse* than the selector it is given (multi-omics -0.045 SD, CI -0.11 to -0.003, Wilcoxon $p = 0.05$; mixed model $p = 0.21$).
 - Without attributions, the KG-only agent is indistinguishable from a random convergent pathway (+0.01 SD, $p = 0.78$).

So the null is upstream of the LLM: there is no dependency signal in the attribution-supported pathways for any agent to find.

4.5 What the agents claimed

The KG-only agent concentrated on fewer pathways (diversity 0.17–0.20) than the attribution agents (0.26–0.86), so it falls back on a few familiar convergent pathways when given no other signal. Mean

Table 6: Decomposition with deterministic, no-LLM controls (synergy stratum). The selector picks the highest-scoring admissible pathway; “random” is the expected score of a uniform choice among convergent pathways.

Comparison	n items	Cell lines	Mean Δ	95% CI	d_z	p_{mixed}	p_{Wilcoxon}
<i>What the attributions add (no LLM)</i>							
Attribution selector – random (multi-omics)	98	33	+0.093	[−0.05, +0.26]	+0.12	0.29	0.62
Attribution selector – random (L1000)	21	18	+0.051	[−0.21, +0.35]	+0.08	0.53	1.00
<i>What the LLM adds on top of the same input</i>							
LLM agent – attribution selector (multi-omics)	97	33	−0.045	[−0.11, −0.00]	−0.16	0.21	0.050
LLM agent – attribution selector (L1000)	21	18	+0.002	[−0.03, +0.04]	+0.03	0.90	0.65
KG-only LLM agent – random	242	41	+0.009	[−0.08, +0.11]	+0.01	0.78	0.62

Table 7: Properties of the claimed top-1 pathways. Common-essential fraction: the share of the pathway’s genes that are DepMap common-essential. Diversity: distinct pathways named divided by items with a claim.

Condition	Stratum	Mean spec (top-1)	Common-essential fraction	Pathway diversity	Median pathway size
Attribution agent (multi-omics)	antagonism	−0.003	0.38	0.26	139
Attribution agent (multi-omics)	synergy	−0.024	0.34	0.33	139
Attribution agent (L1000)	antagonism	−0.046	0.23	0.78	183
Attribution agent (L1000)	synergy	+0.046	0.28	0.86	181
KG-only agent	antagonism	+0.081	0.33	0.17	139
KG-only agent	synergy	−0.018	0.26	0.20	160
Attribution selector (multi-omics)	antagonism	+0.031	0.37	0.26	139
Attribution selector (multi-omics)	synergy	+0.011	0.35	0.31	139
Attribution selector (L1000)	antagonism	−0.046	0.23	0.78	183
Attribution selector (L1000)	synergy	+0.044	0.28	0.81	181
Random convergent pathway	antagonism	+0.050	0.32	–	139
Random convergent pathway	synergy	−0.032	0.27	–	139

specificity of the claimed pathway is near zero in every condition. Their common-essential fractions are similar (0.23–0.38), so neither condition simply picks pathways that are essential everywhere (Table 7).

5 Interpretation

- **The KG constraint works; the attributions do not inform.** The design goal of preventing hallucinated mechanisms was met: almost every claim is a verified KG path, and the agent abstains when nothing is admissible. But a constrained explanation is only as informative as its constraint. Attribution-supported pathways are no better at locating cell-line-specific dependencies than random convergent pathways.
- **Why attributions carry no mechanism here.**
 - SynergyLM sees 59 distinct cell profiles through 27,606 omics features. Dropping any one modality leaves test r unchanged (0.764–0.765), and dropping all three costs 0.21.
 - The follow-up study (SynergyLM-v2) tested this directly: replacing the omics encoder by a one-hot cell embedding makes no difference even for unseen drugs (median per-drug $\Delta r = -0.008$, $p = 0.48$).
 - Reactome pathway features of drug targets perform no better than a structure-matched random-pathway control ($\Delta r = +0.002$, $p = 1$).

The omics act as a cell-line identifier, so gene-level attributions describe *which* cell line it is rather than *why* the pair synergizes.

- **The expression confound (threat T1) was visible early.** In a pre-run smoke test on two items, the attribution-supported pathways overlapped substantially (Jaccard 0.27 and 0.48) with

the pathways enriched when the “attribution” was just the cell line’s expression deviation from the mean profile. Attribution support is therefore partly expression-driven.

- **Implication for the extension.** A larger or paid LLM would not rescue H1, because the deterministic selector shows the signal is absent before any LLM sees it. Pathway-level explanations would need a model whose cell-side representation is not reducible to identity, or ground truth tied to the combination rather than to single-gene dependency (threat T4).

6 Limitations and threats to validity

- **Power.** Complete-case H1 has 93 (multi-omics) and 20 (L1000) items rather than the planned 240. The multi-omics null is informative (CI upper bound +0.23 SD); the L1000 null is not.
- **Ground truth.** Single-gene CRISPR dependency of a pathway is a proxy: a pathway can be essential in a line without mediating the combination (T4).
- **Coverage.** 41 of 59 cell lines have DepMap CRISPR data, and 29 of 129 drugs have no protein target (mostly cytotoxics), so the claim is restricted to targeted agents in covered lines.
- **One LLM.** The study used one open-weight model. The decomposition argues that model choice is not the limiting factor, but other LLMs were not run.
- **L1000 signatures** are consensus profiles from other cell lines, not the DrugComb line (T6).

Reproducibility

- **Code:** branch `synergylm-v2`, commit `93de460` (“attribution-guided pathway explanation study”). Entry point `explain_pathways.py` with stages `download`, `labels`, `map`, `kg`, `predict`, `sample`, `attribute`, `pathways`, `select`, `agent`, `score`, `analyze`, `gates`. Library: `src/explain/`. Tests: `tests/test_explain.py`.
- **Jobs:** `run_xai_train.sh` (models), `run_xai_pipeline.sh` (attributions, pathways, G0/G1), and `run_xai_agent.sh`. The last serves the LLM with vLLM, runs the 20-item pilot and gate G2, then the full agent run, scoring and analysis.
- **Results:** `data/xai/results/analysis.json` and `scored.csv` (3,000 rows: 6 conditions $\times$ 500 items). Agent transcripts are in `data/xai/agent/`, and the failed pilot 1 is archived in `data/xai/agent_archive/`.
- **This report:** `reports/llm_interpretation/`. `make_tables.py` regenerates every table and the figure from `analysis.json`.